

Vinícius de Paulo Souza Ribeiro

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ACADEMIC EDUCATION

Doctor of Philosophy (Ph.D.) – Université de Lorraine

Oct 2020 – Dec 2023

Computer Science – Deep Learning, Speech Production

Thesis title: *Deep Supervision of the Vocal Tract Shape for Articulatory Synthesis of Speech*

Extracurricular activities:

- 12th Lisbon Machine Learning School – Instituto Superior Técnico, Lisboa, Portugal (July 2022)
- 4th International School on Deep Learning – Università Politecnica delle Marche, Universitat Rovira i Virgili & IRDTA, Las Palmas de Gran Canaria, Spain (July 2021)

Master of Science (M. Sc.) – University of Campinas

Aug 2017 – Aug 2019

Electrical Engineering – Deep Learning, Computer Vision

Dissertation title: *The Impact of Annotation Quality on Deep Learning for Skin Lesion Segmentation*

Extracurricular activities:

- Summer School on Data Science for Document Analysis and Understanding – La Rochelle Université & Inria, France (July 2019)

Bachelor of Science (B. Sc.) – University of Campinas

Feb 2011 – Aug 2016

Electrical Engineering

Extracurricular activities:

- Software Engineering Internship – Motorola Mobility, Jaguariúna, Brazil (2015 – 2016)
- Linear Analysis Lab Teaching Assistant – University of Campinas, Campinas, Brazil (1st Semester, 2015)
- Academic Internship – Czech Technical University, Prague, Czech Republic (Jan – Feb 2015)
- Academic Internship – University of Campinas, Campinas, Brazil – Funded by PIBITI/CNPq (2014 – 2015)

LANGUAGES

Portuguese	Native
English	Fluent
French	Advanced

PROFESSIONAL EXPERIENCES

The following is a selection of my most recent positions. For a complete work history, visit my [LinkedIn](#) profile.

Fin AI

Jan 2024 – Current

Senior Machine Learning Scientist

Dublin, Ireland

- Fine-tune open-source large language models via supervised fine-tuning and reinforcement learning, with a focus on building production-ready agentic AI capabilities
- Architect AI agent systems; drive model quality through iterative prompt engineering and rigorous A/B experimentation

Loria*Doctoral Researcher (Ph.D. Candidate)*

Oct 2020 – Dec 2023

Nancy, France

- Led deep learning research on articulatory speech synthesis, developing neural network models trained on multimodal physiological and acoustic data
- Built models for real-time MRI segmentation, articulatory motion tracking, and speech processing
- Published peer-reviewed, state-of-the-art research at top international conferences and in high-impact journals

Meta AI*Research Scientist Intern*

Aug 2022 – Dec 2022

New York, USA

- Collaborated with Meta AI's Speech Team to investigate alignment techniques for robust wake word detection
- Delivered research resulting in a first-author publication at Interspeech 2023

Dasa & Nexa Digital*Data Science Specialist*

Oct 2018 – Oct 2020

São Paulo, Brazil

- Developed deep learning models for medical image analysis across CT, MRI, ultrasound, and X-ray modalities
- Built ML solutions that streamlined clinical workflows and reduced manual effort for medical teams
- Designed ETL pipelines to extract structured information from unstructured PDF medical reports
- Established data governance and traceability standards across ingestion, processing, and storage workflows

PUBLICATIONS

My work has appeared in premier international venues including *CVPR*, *Interspeech*, and *ICA*, as well as high-impact journals such as *Speech Communication* and *Computer Methods and Programs in Biomedicine*, with additional pre-print work on arXiv.

Below is a short list of relevant publications in which I am the first author. For my complete list of publications, visit my [Google Scholar](#) profile or my [personal website](#).

Automatic Segmentation of Vocal Tract Articulators in Real-Time Magnetic Resonance Imaging 2023*Vinicius Ribeiro, Karyna Isaieva, Justine Leclere, Jacques Felblinger, Pierre-André Vuissoz, Yves Laprie*

Computer Methods and Programs in Biomedicine

Autoencoder-Based Tongue Shape Estimation During Continuous Speech 2022*Vinicius Ribeiro, Yves Laprie*

Interspeech 2022

Automatic Generation of the Complete Vocal Tract Shape from the Sequence of Phonemes to be Articulated 2022*Vinicius Ribeiro, Pierre-André Vuissoz, Karyna Isaieva, Justine Leclere, Yves Laprie*

Speech Communication

Towards the Prediction of the Vocal Tract Shape from the Sequence of Phonemes to be Articulated 2021*Vinicius Ribeiro, Karyna Isaieva, Justine Leclere, Pierre-André Vuissoz, Yves Laprie*

Interspeech 2021